

Paleolithic vs. modern diets--selected pathophysiological implications.



The nutritional patterns of Paleolithic humans influenced genetic evolution during the time segment within which defining characteristics of contemporary humans were selected. Our genome can have changed little since the beginnings of agriculture, so, genetically, humans remain Stone Agers--adapted for a Paleolithic dietary regimen. Such diets were based chiefly on wild game, fish and uncultivated plant foods. They provided abundant protein; a fat profile much different from that of affluent Western nations; high fibre; carbohydrate from fruits and vegetables (and some honey) but not from cereals, refined sugars and dairy products; high levels of micronutrients and probably of phytochemicals as well. Differences between contemporary and ancestral diets have many pathophysiological implications. This review addresses phytochemicals and cancer; calcium, physical exertion, bone mineral density and bone structural geometry; dietary protein, potassium, renal acid secretion and urinary calcium loss; and finally sarcopenia, adiposity, insulin receptors and insulin resistance. While not, yet, a basis for formal recommendations, awareness of Paleolithic nutritional patterns should generate novel, testable hypotheses grounded in evolutionary theory and it should dispel complacency regarding currently accepted nutritional tenets.